

Modelling with hyperbolic systems of conservation laws

SEAMS school 2026 “Applying mathematics to real-world problems”

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“Mathematics is everywhere”

That is true and we could *prove it*.

Issue(s): either society or students do not feel it. Moreover students do like mathematics less and less.

Approach strategies:

- Family: importance of **STEM**, salaries are higher than. . .
- Society: relevance in solving real-world problems, from technical as AI to medicine and health sciences ones.
- Teachers: change of paradigma and new strategies. . . young people likes computers and mobiles— their best friends—, e.g., **AI** is at fashion and it is full founded in mathematics, statistics and computer science.
- Students: connexion with real-world, mathematics comes naturally (demythification), it is useful: i) a language, ii) a tool, iii) a science), in fact it is iv) a native language and v) a *meta-science*.

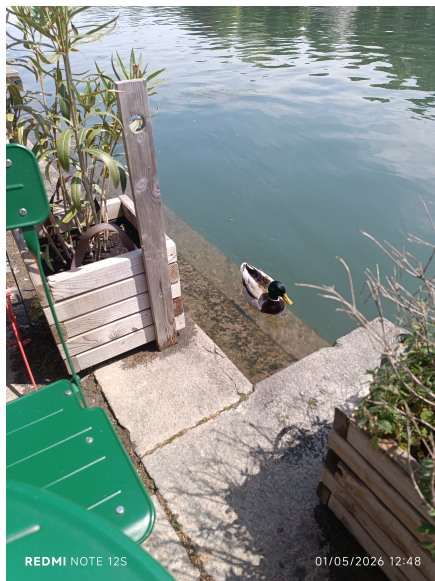
Modelling¹ with hyperbolic systems of conservation laws (lectures original plan)

Abstract. In this course **real-world problems** provide **motivation** for the mathematical **concepts/theory** of hyperbolic systems of conservation laws. In particular we will consider a few examples presented at some **ESGI** (European Study Groups with Industry) meetings. Here is a list of topics under consideration:

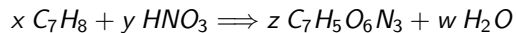
- 1 Euler's compressible gas dynamics (first order conservation laws): shock waves, Rankine-Hugoniot condition, hyperbolicity, and entropy criteria;
- 2 pollutant effects on stone monuments (convective, diffusive and capillary mechanisms): porous-capillary media;
- 3 image processing (denoising and segmentation): nonlinear diffusions;
- 4 reacting gas flows or coagulation-fragmentation theory: approximations and the viscosity-capillary vanishing method.

¹“**keywords**” in red colour

Giuseppe Luigi Lagrange Dept. Math. Sci., PoliTo, Turin, Italy: “Quo vadis? (or Doing maths as a duck).”



Lavoisier's conservation law



Euler's gas dynamics equations

Compressible, inviscid, no heat conduction, in Eulerian coordinates:

$$\rho_t + \sum_{j=1}^3 (\rho v_j)_{r_j} = 0$$

$$(\rho v_i)_t + \sum_{j=1}^3 (\rho v_i v_j + \delta_j^i p)_{r_j} = 0, \quad i = 1, 2, 3$$

$$(\rho E)_t + \sum_{j=1}^3 ((\rho E + p) v_j)_{r_j} = 0$$

$$p = p(\rho, e), \quad E = \frac{1}{2} |v|^2 + e$$

ρ is density, v is velocity, p is pressure, e is internal energy and E is total energy.

Shocks



A MANUAL ON THE MATHEMATICAL THEORY
OF NON-LINEAR WAVE MOTION

APPLIED MATHEMATICS PANEL
NATIONAL DEFENSE RESEARCH COMMITTEE

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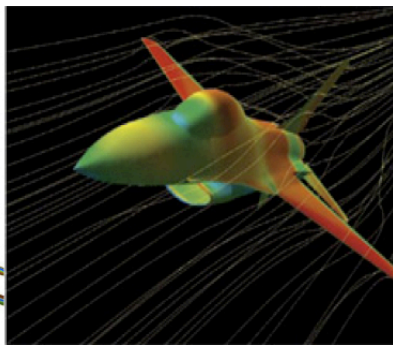
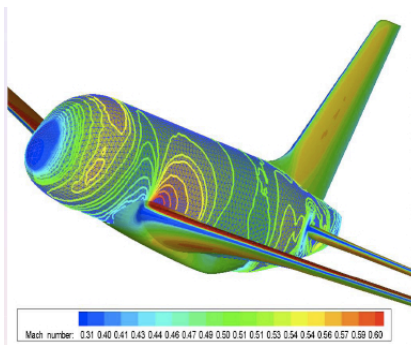
R. Courant
K.O. Friedrichs

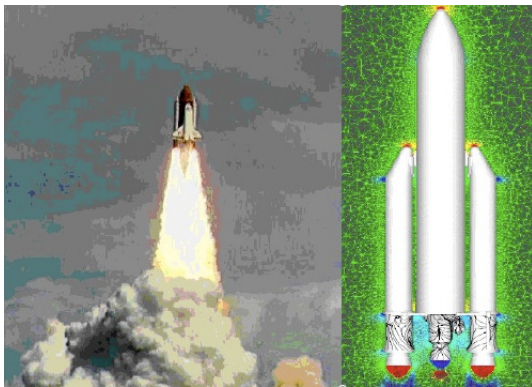
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Supersonic Flow and Shock Waves

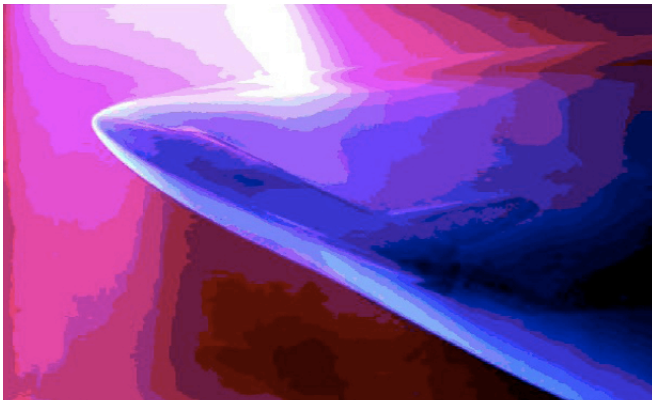


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NASA: Hypersonic flow with shocks at the nose



NASA: Visible shocks at the nose in the windtunnel test

Optimal design and **active control** in structures: bridges, the Thames barrier, wind towers, aeroplanes or the shuttle and the orbital spatial station...

... **shocks** and **oscillations** are fundamental issues:

Multivalued functions [Ris] p. 6, 7

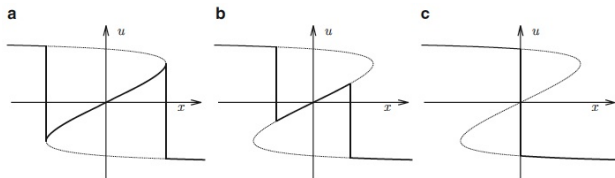
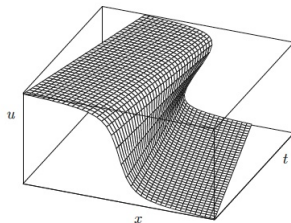


Fig. 1.2 Different solutions with u conserved

Population balance equations

Euler's Gas Dyn. Eqs.

+

$$\frac{\partial f_1(\mathbf{z}, t; \mathbf{y})}{\partial t} + \nabla \cdot (\dot{\mathbf{Z}}(\mathbf{z}, t; \mathbf{y}) f_1(\mathbf{z}, t; \mathbf{y})) = h(\mathbf{z}, t; \mathbf{y})$$

$\mathbf{z} = (\mathbf{x}, \mathbf{r}) \in \Omega_{\mathbf{x}} \times \Omega_{\mathbf{r}}$ the state internal variables $\mathbf{x}$ and external (space) variables $\mathbf{r}$

$\mathbf{y} = \mathbf{y}(\mathbf{r}, t) = (\rho, \rho \mathbf{v}, \rho E)$ the phase variables

$\dot{\mathbf{Z}} = (\dot{\mathbf{X}}(\mathbf{z}, t, \mathbf{y}), \dot{\mathbf{R}}(\mathbf{z}, t, \mathbf{y}))$ the velocity field

f_1 the number density, h the “source”

Table 1 Applications of population balance modeling in different areas

Areas of applications	Specific details	References
Agriculture engineering	Fractionation of ground switchgrass, hydrolysis of cellulose, airborne dust removal	1–4
Astrophysics and astronomy	Population balance equations for levels of a quantum system, calculation of highly excited hydrogen b_n factor	5, 6
Biochemistry and molecular biology	Gene regulatory processes, transcription, translation	7–13
Biomedical engineering	Kidney stone formation, cancer treatment	14, 15
Biotechnology and bioengineering	Cell cycle dynamics, cellulase production in airlift reactor, bioreactor design, cell proliferation, cell morphology, aeration of bread dough, growth of hairy roots, enzymatic lysis of cells	16–27
Cell biology	Cell aggregation dynamics, cryopreservation, stem cell differentiation, mammalian cell culture, neutropenia	28–36
Civil and environment engineering	Water softening, deepwater plumes, dissolved air flotation, struvite crystallization in urine, activated sludge flocculation, drag reduction	37–42
Electrochemistry	Bubble dynamics at gas-evolving electrodes, chemical vapor deposition, electrostatic precipitators	43–45
Energy fuels	Gas-liquid-solid-phase reactor, silicon nanoparticles, oxy-coal jet flames, hydrolysis of cellulose, Fischer-Tropsch synthesis, asphaltene precipitation, reactive extractions, pyrolysis of wood, coal combustion, enhanced oil recovery	3, 46–54

Geography	Electrification of particles in dust storms, flocculation of colloidal montmorillonite, soil respiration	55–57
Hematology	Kinetics of platelet aggregation, leukemic cell proliferation	36, 58
Novel functional particles	Single-walled carbon nanotube formation	59
Ophthalmology	Pterygium formation, corneal epithelial maintenance and graft losses	60, 61
Personalized medicine	Chemotherapy for leukemia	15, 58
Pharmaceutical engineering	Polymorphic transformation, crystal morphology distribution, nucleation, Ostwald ripening, continuous crystallization, dissolution, breakage, aggregation, granulation, tablet manufacturing	62–72
Respiratory systems	Modeling techniques for the deposition of inhaled particles	73
Telecommunications	Impact evaluation of rare metals in waste mobile phone and personal computer	74
Transportation	Soot particle-size distribution emitted from diesel engine, soot formation, CO ₂ emission	75–77

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²D. Ramkrishna, M. R. Singh, Population Balance Modeling: Current Status and Future Prospects, Annu. Rev. Chem. Biomol. Eng. 2014.5:123-146